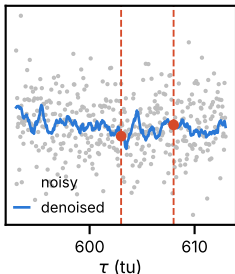


(a) Input window
4 states $\times$ 2 times



(f) Classical menu

linear ridge
NG-RC (degree 2, 3)
random features
(dimension-matched)

same window,
targets and readout

(b) Local-detuning
encoding (rad/ μ s)

$$\Delta_i = 4.5 - 9 x_i$$

$$a = 10 \mu\text{m}$$

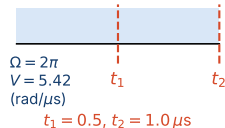
blue $\Delta_i < 0$, red $\Delta_i > 0$

sign-definite variant:
 $\Delta_i = -0.5 - 9 x_i$ (Sec. 4.3)



(c) Quench from $|g\dots g\rangle$,
one run per probe time

$$H = \sum_i \left[\frac{\Omega}{2} \sigma_x^{(i)} - \Delta_i n_i \right] + \sum_{i < j} V_{ij} n_i n_j$$



(e) Readout:
increment prediction

$$\hat{\mathbf{y}} = \mathbf{W}\mathbf{E} + \mathbf{b}$$

ridge, penalty chosen by
CV on training anchors

$$\mathbf{y} = \mathbf{x}(\tau + H) - \mathbf{x}(\tau)$$

$$H \in \{1, 2, 3\} \tau_c$$

(d) Embedding $E \in \mathbb{R}^{16}$
(Bloqade, 500 shots)

